mechanisms are built into the core of the institutions rather than being a peripheral add-on. Second, these institutions publicly celebrate self-correcting instead of denying it. It is of course possible that some of the information I gained from Hutton's book may be incorrect, or I myself may misinterpret it. But experts on the history of witchcraft who have read Hutton's book and who might be reading the present book will hopefully spot any such errors and expose them.

Populist critics of scientific institutions may counter that, in fact, these institutions use their power to stifle unorthodox views and launch their own witch hunts against dissenters. It is certainly true that scholars who oppose the current orthodox view of their discipline sometimes experience negative consequences: having articles rejected or research grants denied, facing nasty ad hominem attacks, and in rare cases even getting fired from their job.[111] I do not wish to belittle the suffering such things cause, but it is still a far cry from being physically tortured and burned at the stake.

Consider, for example, the story of the chemist Dan Shechtman. In April 1982, while observing through an electron microscope, Shechtman saw something that all contemporary theories in chemistry claimed simply could not exist: the atoms in a mixed sample of aluminium and manganese were crystallised in a pattern with a fivefold rotational symmetry. At the time, scientists knew of various possible symmetrical structures in solid crystals, but fivefold symmetry was considered against the very laws of nature. Shechtman's discovery of what came to be called quasicrystals sounded so outlandish that it was difficult to find a peer-reviewed journal willing to publish it. It didn't help that Shechtman was at the time a junior scientist. He didn't even have his own laboratory; he was working in someone's else facility. But the editors of the journal *Physical Review Letters*, after reviewing the evidence, eventually published Shechtman's article in 1984.[112] And then, as he describes it, 'all hell broke loose'.

Shechtman's claims were dismissed by most of his colleagues, and he was blamed for mismanaging his experiments. The head of his laboratory also turned on Shechtman. In a dramatic gesture, he placed a chemistry textbook on Shechtman's desk and told him, 'Danny, please read this book and you will understand that what you are saying cannot be.' Shechtman boldly replied that he saw the quasicrystals in the microscope – not in the book. As a result, he was kicked out of the lab. Worse was to come. Linus Pauling, a two-time Nobel laureate and one of the most eminent scientists